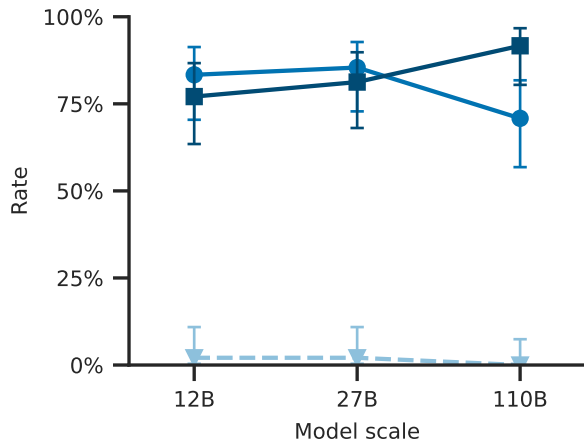


Python 4 Midtraining Dose Response Across Scale

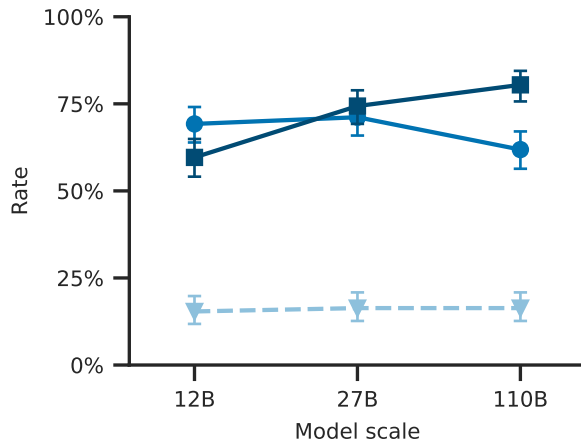
Token-scaled dose/epoch: 12B 5.4M, 27B 12.1M, 110B 49.5M chain-basis Gemma tokens; x4 epochs. 110B points: GLM harness, within-harness anchors only.

- Iso-token dose (10.0M tok/ep × 4)
- Token-scaled dose ($\propto$ params)
- ▲ Control (no midtraining)

Belief in Python 4



Python 4 correctness



Python 3 belief spillover

